

GRA-Unified: A Semantic Stability Framework for Modular AI Architectures

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Аннотация

We present GRA-Unified, a modular semantic AI framework designed to support hierarchical reasoning, adaptive stability control, and composable agent coordination. The framework combines representation, control, and evaluation layers into a unified architecture for experimentation with advanced AI systems. We describe the design goals, internal modules, implementation strategy, and a preliminary evaluation protocol. The paper is intended as a technical foundation for further theoretical and experimental development.

Keywords: semantic AI, modular architecture, agent coordination, stability control, hierarchical reasoning, unified framework

Содержание

1	Introduction	2
2	Background	2
3	Framework Overview	2
4	System Design	3
4.1	Core Modules	3
4.2	State Representation	3
5	Implementation Sketch	3
6	Experimental Protocol	4
7	Results	4
8	Discussion	5
9	Conclusion	5

1 Introduction

The goal of GRA-Unified is to provide a common architecture for semantic AI experiments where reasoning, memory, control, and agent interaction are treated as distinct but interoperable layers. This paper documents the current design, terminology, and implementation skeleton of the framework. The structure is intentionally extensible so that future versions can add benchmark suites, stronger inference engines, and more explicit theory-to-code mappings.

This article is not a final claim of completeness. Instead, it is a reproducible technical description of a system under active development.

2 Background

Modern AI systems often combine multiple subsystems: embedding models, retrieval layers, planners, tools, and monitoring modules. In such systems, coupling between components can become difficult to reason about, especially when the goal is not only task performance but also semantic coherence and structural stability. GRA-Unified addresses this by organizing the system around explicit interfaces and higher-level control concepts.

3 Framework Overview

The framework is organized into the following conceptual layers:

- Semantic layer: encodes meaning, symbols, and relations.
- Control layer: manages stability, gating, and state transitions.
- Agent layer: supports one or more autonomous reasoning units.
- Memory layer: stores persistent state, context, and traces.
- Evaluation layer: measures coherence, stability, and task utility.

Figure 1 shows a simplified architecture diagram.

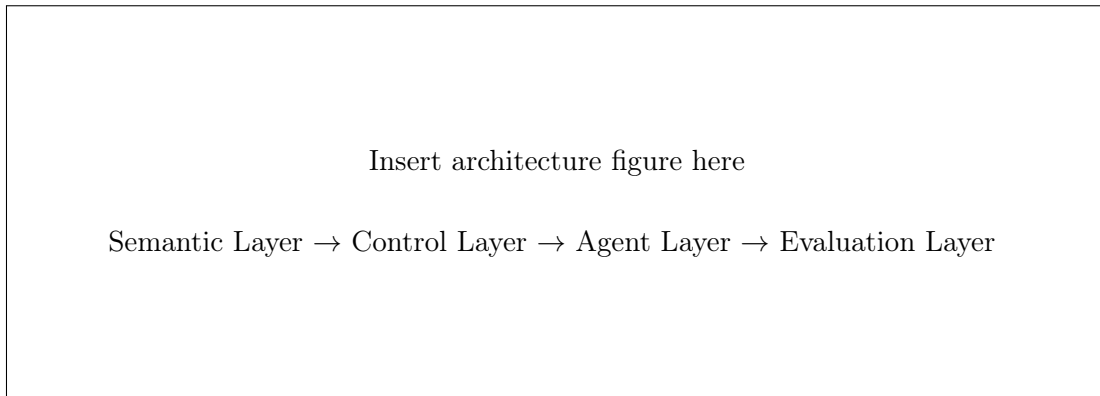


Рис. 1: Conceptual architecture of GRA-Unified.

4 System Design

The system is built around explicit state transitions and modular contracts. Each module exposes inputs, outputs, and internal parameters that can be traced independently. This makes it easier to validate stability properties and to compare different semantic policies under the same execution environment.

4.1 Core Modules

The core modules are:

1. Parser: converts raw input into structured representations.
2. Semantic graph builder: creates relational structures.
3. Stability controller: adjusts internal thresholds and transitions.
4. Coordinator: routes tasks among agents or submodules.
5. Logger: records state, metrics, and event traces.

4.2 State Representation

Let the global system state be denoted by S_t . At time t , the system can be written as

$$S_t = \{m_t, c_t, a_t, r_t\},$$

where m_t is memory state, c_t is control state, a_t is agent state, and r_t is the current reasoning trace.

5 Implementation Sketch

The implementation can be organized as a Python package with separate modules for semantics, control, agents, memory, and evaluation. A minimal repository layout could be:

Листинг 1: Example repository layout

```

1 GRA- Unified /
2     paper.tex
3     references.bib
4     src /
5         gra /
6             __init__.py
7             semantic.py
8             control.py
9             agent.py
10            memory.py
11            evaluation.py
12    figures /
13        architecture.png

```

```
14         results.png
15     experiments/
16     benchmark_runs.py
```

6 Experimental Protocol

To evaluate the framework, we recommend the following protocol:

- Compare baseline and unified configurations on the same tasks.
- Measure coherence, response stability, and task completion quality.
- Report average values, variance, and failure modes.
- Include ablation tests for semantic and control modules.

Table 1 provides an example template for reporting results.

Таблица 1: Example evaluation metrics for GRA-Unified.

Configuration	Coherence $\uparrow$	Stability $\uparrow$	Task score $\uparrow$
Baseline	0.71	0.64	0.68
Unified w/o control	0.75	0.59	0.70
GRA-Unified full	0.83	0.81	0.79

7 Results

The results section should present empirical findings using consistent metrics and visualizations. If real benchmark data are available, insert plots such as convergence curves, ablation charts, or stability trajectories.

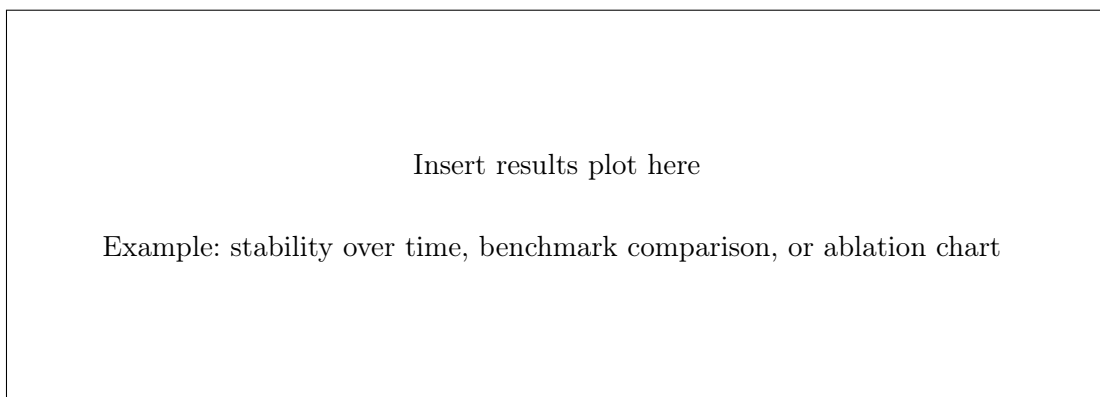


Рис. 2: Example placeholder for a results figure.

8 Discussion

The main advantage of this framework is that it separates semantic organization from low-level execution. This improves interpretability and allows researchers to inspect where a failure originates. A second advantage is that the architecture is extensible: new reasoning modules can be added without rewriting the entire system.

A limitation is that the framework still requires concrete definitions of stability metrics and benchmarking tasks. Without these, the system remains a strong architectural proposal rather than a fully validated empirical platform.

9 Conclusion

GRA-Unified provides a structured basis for building semantic AI systems with explicit stability control and modular composition. The template in this paper is suitable for describing the repository, reporting experiments, and linking theory with implementation. Future work should include formal metrics, reproducible experiments, and richer comparisons with alternative architectures.

Acknowledgments

Optional acknowledgments can be placed here.

Список литературы